Why a Planar Simulation Uses the Three-Dimensional Ferrofluid Equations Without Modification

With notes on the magnetisable medium and on the limitations
of the model of Bacri *et al.* (1995)

What this note shows. If the flow is planar and the magnetic field lies in that same plane, then the out-of-plane component of the magnetisation can never grow: every term in the governing equations has zero z-component. The planar computation is therefore not an approximation of the three-dimensional problem — it *is* the three-dimensional problem, with components that are identically zero simply not stored. No equation and no coefficient needs re-deriving. The note then quantifies what changes once the fluid's own magnetisation is accounted for, and lists eleven approximations that the underlying model carries.

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1. The question, and the answer in one page

1.1 Why the question arises

Bacri's experiment used a cylindrical capillary — a three-dimensional apparatus. The simulation is a planar channel with two-dimensional flow. It is natural to worry that a magnetisation equation written for three dimensions needs re-deriving for two.

The worry comes from one word doing two jobs.

The flow is two-dimensional. The velocity lies in a plane, $\mathbf{v} = (v_x, v_y, 0)$, nothing varies along z, and the vorticity $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \mathbf{v}$ points out of the plane.

The particle orientations are three-dimensional. Each nanoparticle carries a magnetic moment whose direction is a point on a sphere. Brownian rotation moves it over that sphere, and averaging the Boltzmann factor $e^{\xi \cos\theta}$ over the sphere is what produces the Langevin function $L(\xi) = \coth \xi - 1/\xi$.

These are different things at different length scales. A flow that happens to be uniform along z does not reach down to a 10 nm particle and forbid it from tumbling out of the plane.

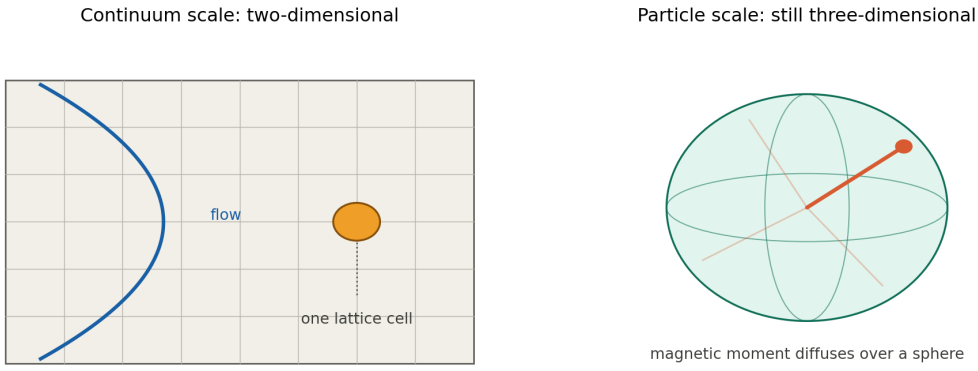


Figure 1: Left: the flow lives on a plane, so a lattice cell is a two-dimensional object. Right: inside that cell are billions of particles whose moments still explore a full sphere. The orientational statistics is a sub-grid closure — a property of the fluid, not of the mesh.

1.2 The answer

The simulation is not a two-dimensional *universe*. It is a three-dimensional system that happens to be uniform along z. We solve the same equations, restricted to a family of states that the equations never leave.

The reason they never leave it is bookkeeping with cross products, and it is worth seeing before the details. With $\mathbf{H}$ in the plane and $\mathbf{\Omega}$ out of it:

$$\mathbf{\Omega}\times\mathbf{M}: \quad \text{(out-of-plane)}\times\text{(in-plane)} \;=\; \text{in-plane}, \tag{1}$$

$$\mathbf{M}\times\mathbf{H}: \quad \text{(in-plane)}\times\text{(in-plane)} \;=\; \text{out-of-plane}, \tag{2}$$

$$\mathbf{M}\times(\mathbf{M}\times\mathbf{H}): \quad \text{(in-plane)}\times\text{(out-of-plane)} \;=\; \text{in-plane}. \tag{3}$$

Every term in the magnetisation equation is one of these three shapes, plus multiples of $\mathbf{M}$ and $\mathbf{H}$ themselves. All stay in the plane. So if M_z starts at zero, $\dot{M}_z$ is zero, and it stays zero forever.

> **Physical statement.** A field lying in the plane exerts a torque about the out-of-plane axis. A torque about z can only spin a particle about z, which keeps its moment in the xy-plane. There is nothing available to tip the moment out of the plane, so nothing does.

2. Setting up

2.1 The equations

We use the system of Bacri *et al.*, Phys. Rev. Lett. **75**, 2128 (1995), their Eqs. (3) and (5)–(6), together with magnetostatics. Momentum:

$$\rho\frac{D\mathbf{v}}{Dt} = -\nabla p + \mu_0(\mathbf{M}\cdot\nabla)\mathbf{H} + \eta\nabla^2\mathbf{v} + \tfrac{1}{2}\mu_0\nabla\times(\mathbf{M}\times\mathbf{H}), \qquad \nabla\cdot\mathbf{v} = 0. \tag{4}$$

Magnetisation, in the effective-field (Martsenyuk–Raikher–Shliomis) form. Here $\boldsymbol{\xi}$ is the true-field Langevin argument, $\boldsymbol{\xi}_e$ the effective field, $\xi_e = |\boldsymbol{\xi}_e|$, $\mathbf{n} = \boldsymbol{\xi}_e/\xi_e$, and $S_2(x) = 1 - 3L(x)/x$:

$$\mathbf{M} = \varphi M_s\, L(\xi_e)\,\mathbf{n}, \tag{5}$$

$$\frac{d}{dt}(L\,\mathbf{n}) = \underbrace{\mathbf{\Omega}\times(L\,\mathbf{n})}_{\text{carried by the flow}} - \underbrace{\frac{1}{\tau_B}\frac{L}{\xi_e}(\boldsymbol{\xi}_e - \boldsymbol{\xi})}_{\text{relaxation toward the field}} - \underbrace{\frac{S_2}{2\tau_B\xi_e^2}\,\boldsymbol{\xi}_e\times(\boldsymbol{\xi}_e\times\boldsymbol{\xi})}_{\text{relaxation is slower across than along}}. \tag{6}$$

Field:

$$\nabla\times\mathbf{H} = 0, \qquad \nabla\cdot(\mathbf{H}+\mathbf{M}) = 0 \qquad\Longrightarrow\qquad \mathbf{H} = \mathbf{H}_{\text{app}} - \nabla\psi, \quad \nabla^2\psi = \nabla\cdot\mathbf{M}. \tag{7}$$

2.2 The four conditions we assume

(C1) Nothing varies along z: $\partial_z(\cdot) = 0$, including the geometry, the boundary conditions and the initial state.

(C2) The flow is planar: $v_z = 0$ initially, and no wall drives flow along z.

(C3) The applied field lies in the plane: $\mathbf{H}_{\text{app}} = (H_x, H_y, 0)$.

(C4) The magnetisation starts in the plane: $M_z = 0$ initially.

These are exactly the simulation as built: a planar channel, uniform along z, with $\mathbf{H}_{\text{app}} = H_0\cos(\omega t)\hat{x}$.

2.3 Two immediate consequences

The vorticity points out of the plane. With $v_z = 0$ and $\partial_z = 0$,

$$\nabla \times \mathbf{v} = \big(\underbrace{\partial_y v_z - \partial_z v_y}_{0},\ \underbrace{\partial_z v_x - \partial_x v_z}_{0},\ \partial_x v_y - \partial_y v_x \big) \implies \boldsymbol{\Omega} = \Omega \hat{z}. \tag{8}$$

This holds in *any* planar flow — around obstacles, in bends, at a free surface. It is not special to Poiseuille flow.

The computed field stays in the plane. In (7) the source is $\nabla\cdot\mathbf{M} = \partial_x M_x + \partial_y M_y$, which has no z-dependence when $M_z = 0$ and $\partial_z = 0$. With z-independent boundary data, ψ is independent of z, so $\nabla\psi = (\partial_x\psi, \partial_y\psi, 0)$ lies in the plane. Since $\mathbf{H}_{\text{app}}$ lies in the plane by (C3), so does $\mathbf{H}$.

This second point matters later: it says the field solver, once we add it back, does not break the planar structure either.

3. The check: no term has a z-component

We now write out every term and read off its z-component. Nothing below is more than substitution.

3.1 The magnetisation equation

Take $\mathbf{M} = (M_x, M_y, 0)$, $\mathbf{H} = (H_x, H_y, 0)$, $\boldsymbol{\Omega} = (0, 0, \Omega)$.

Term	**Written out**	**Its z-component**
$\boldsymbol{\Omega} \times \mathbf{M}$	$\big(-\Omega M_y,\ \Omega M_x,\ 0 \big)$	zero: $\hat{z}\times$ anything in the plane stays in the plane
$\mathbf{M} \times \mathbf{H}$	$\big(0,\ 0,\ M_x H_y - M_y H_x \big)$	*purely* z — this is the torque
$\mathbf{M} \times (\mathbf{M} \times \mathbf{H})$	$\big(-M_y T,\ M_x T,\ 0 \big),\ T = M_x H_y - M_y H_x$	zero, by the line above
$\boldsymbol{\xi}_e - \boldsymbol{\xi}$	$\big(\xi_{e,x} - \xi_x,\ \xi_{e,y} - \xi_y,\ 0 \big)$	zero: difference of two in-plane vectors
$\mathbf{M}_0 \propto \mathbf{H}$	$\big(\propto H_x,\ \propto H_y,\ 0 \big)$	zero: parallel to $\mathbf{H}$

The third line deserves a pause. Using $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b})$,

$$\mathbf{M} \times (\mathbf{M} \times \mathbf{H}) = \mathbf{M}\,(\mathbf{M}\cdot\mathbf{H}) - \mathbf{H}\,|\mathbf{M}|^2, \tag{9}$$

whose z-component is $M_z(\mathbf{M}\cdot\mathbf{H}) - H_z|\mathbf{M}|^2$. Both pieces vanish: the first because $M_z = 0$, the second because $H_z = 0$. *Both* are needed — which is exactly why condition (C3) cannot be dropped (§6).

Every term of (6) appears in this table, so

$$\boxed{\dot{M}_z = 0 \quad \text{whenever} \quad M_z = 0} \tag{10}$$

3.2 The momentum equation

The same exercise for (4), to confirm the *flow* also stays planar:

Term	**Why its z-component vanishes**
$-\nabla p$	p does not depend on z, so $\partial_z p = 0$.
$\eta\nabla^2\mathbf{v}$	∇^2 acts on each component separately, and $v_z \equiv 0$.
$\rho(\mathbf{v}\cdot\nabla)\mathbf{v}$	$\mathbf{v}\cdot\nabla = v_x\partial_x + v_y\partial_y$, a scalar operator applied to an in-plane vector.
$\mu_0(\mathbf{M}\cdot\nabla)\mathbf{H}$	Same reason: $M_x\partial_x + M_y\partial_y$ applied to $\mathbf{H}$, which is in-plane.
$\frac{1}{2}\mu_0\nabla\times(\mathbf{M}\times\mathbf{H})$	$\mathbf{M}\times\mathbf{H} = T\hat{z}$ from the table above, and $\nabla\times(T\hat{z}) = (\partial_y T,\, -\partial_x T,\, 0)$.

So $\dot{v}_z = 0$: a flow that starts planar stays planar. Together with §3.1 and the field result of §2.3, the whole coupled system — flow, magnetisation and field — stays in the plane.

3.3 Why "it stays planar" settles the matter

One might object: we have shown that a planar solution exists, but could the real three-dimensional fluid do something else and drift out of the plane anyway?

It cannot, because at each point the magnetisation equation is an ordinary differential equation with a *smooth* right-hand side, and a smooth ODE has exactly one solution for given initial data. There is no second solution available for the fluid to choose.

The only place where smoothness is not obvious is $\xi_e \to 0$, where the factors $1/\xi_e$ and $1/\xi_e^2$ in (6) look as if they blow up. They do not. Regrouping (6),

$$\frac{d\boldsymbol{\xi}_e}{dt} = \Omega\,(\hat{z}\times\boldsymbol{\xi}_e) - A\,\boldsymbol{\xi}_e + c_\perp\,\boldsymbol{\xi} + B\,(\boldsymbol{\xi}_e\cdot\boldsymbol{\xi})\,\boldsymbol{\xi}_e, \tag{11}$$

with, writing $x = \xi_e$,

$$A = \frac{L(x)}{x\,L'(x)\,\tau_B}, \qquad c_\perp = \frac{1}{2\tau_B}\left(\frac{x}{L(x)} - 1\right), \qquad B = \frac{A - c_\perp}{x^2}. \tag{12}$$

Expanding with $L = \frac{x}{3} - \frac{x^3}{45} + \cdots$ and $L' = \frac{1}{3} - \frac{x^2}{15} + \cdots$:

$$A\,\tau_B = 1 + \tfrac{2}{15}x^2 + O(x^4), \qquad c_\perp\tau_B = 1 + \tfrac{1}{10}x^2 + O(x^4), \tag{13}$$

so the difference starts at second order,

$$(A - c_\perp)\,\tau_B = \left(\tfrac{2}{15} - \tfrac{1}{10}\right)x^2 + O(x^4) = \tfrac{1}{30}x^2 + O(x^4), \qquad \text{hence} \qquad B\,\tau_B = \tfrac{1}{30} + O(x^2). \tag{14}$$

The x^2 in the numerator cancels the x^2 in the denominator exactly. All three coefficients are finite at $x = 0$, and each is a series in $x^2 = \boldsymbol{\xi}_e\cdot\boldsymbol{\xi}_e$ — so they are smooth functions of the components of $\boldsymbol{\xi}_e$, with no kink at the origin. In the limit,

$$\frac{d\boldsymbol{\xi}_e}{dt} \;\longrightarrow\; \Omega\,(\hat{z}\times\boldsymbol{\xi}_e) - \frac{\boldsymbol{\xi}_e - \boldsymbol{\xi}}{\tau_B}, \tag{15}$$

which is ordinary linear relaxation. This is also why the code needs no special case at $\xi_e = 0$ beyond a guard against dividing by exactly zero.

Conclusion. The planar states are preserved by every equation, and the solution is unique, so the planar solution is *the* solution. The two-dimensional computation reproduces the three-dimensional dynamics exactly. Nothing is neglected, no coefficient is changed, and no error is introduced by the reduction.

3.4 The equations actually solved

Splitting (6) into the part along $\mathbf{n}$ and the part across it gives the form used in the code:

$$\frac{d\boldsymbol{\xi}_e}{dt} = \Omega(\hat{z}\times\boldsymbol{\xi}_e) - \frac{L(\xi_e)}{L'(\xi_e)\tau_B}\left(1-\frac{\mathbf{n}\cdot\boldsymbol{\xi}}{\xi_e}\right)\mathbf{n} + \frac{1}{2\tau_B}\left(\frac{\xi_e}{L(\xi_e)}-1\right)\Big[\boldsymbol{\xi}-\mathbf{n}(\mathbf{n}\cdot\boldsymbol{\xi})\Big], \tag{16}$$

with $\mathbf{M}=\varphi M_s L(\xi_e)\mathbf{n}$, and the magnetic body force

$$\mathbf{F}_m = \mu_0(\mathbf{M}\cdot\nabla)\mathbf{H} + \tfrac{1}{2}\mu_0\big(\partial_y T,\ -\partial_x T,\ 0\big), \qquad T = M_x H_y - M_y H_x. \tag{17}$$

4. Cross-check against Bacri's own equations

Equation (6) of the source paper is poorly legible in the published text, so rather than trusting a reading of it we verify (16) by recovering the two equations Bacri actually integrate to produce Fig. 3. As a by-product this settles the ambiguous coefficient as $S_2 = 1-3L/\xi_e$.

Step 1: switch the flow off. With $\Omega=0$ the effective field lies along $\hat{x}$, the transverse term vanishes, and (16) collapses to

$$\tau_B\frac{d\xi_e}{dt} = -\frac{L}{L'}\left(1-\frac{\xi_0\cos\omega t}{\xi_e}\right) = -\left(\frac{d\ln L}{d\xi_e}\right)^{-1}\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right), \tag{18}$$

which is Bacri Eq. (7).

Step 2: switch the flow back on, weakly. Put $\boldsymbol{\xi}_e = \xi_e\hat{x}+\delta\hat{y}$ with $\delta = \xi_e F\tau_B\Omega$ and $\xi_t = \xi_0\cos\omega t$. To first order $\mathbf{n} = \hat{x}+(\delta/\xi_e)\hat{y}$ and the transverse field is $\boldsymbol{\xi}_\perp = -(\delta\xi_t/\xi_e)\hat{y}$, so the $\hat{y}$-component of (16) is

$$\dot{\delta} = \Omega\xi_e + \frac{\delta\,\dot{\xi}_e}{\xi_e} - \frac{\xi_t\,\delta}{2\tau_B\,\xi_e}\left(\frac{\xi_e}{L}-1\right). \tag{19}$$

Substituting $\delta=\xi_e F\tau_B\Omega$, the terms containing $\dot{\xi}_e$ cancel, and dividing through by $\Omega\xi_e$ leaves

$$\tau_B\frac{dF}{dt} = 1-\frac{1}{2}\left(\frac{1}{L}-\frac{1}{\xi_e}\right)\xi_0 F\cos\omega t, \tag{20}$$

which is Bacri Eq. (8) exactly, including the grouping $(1/L-1/\xi_e)$.

5. The pipe and the slab

Bacri's capillary is cylindrical. Three reasons this does not matter.

(a) Fig. 3 is theory, not experiment. It is produced by integrating Eqs. (7)–(8), which describe a single particle in a local field and a local vorticity. No pipe radius, no channel height and no wall appears in them. They are a *constitutive law* — a statement about the material — and one does not re-derive the viscosity of water for a pipe as opposed to a slab.

(b) The pipe is the planar problem, wrapped. In the capillary the field is axial, $\mathbf{H}=H\hat{x}$, and the vorticity of pipe flow is azimuthal, $\boldsymbol{\Omega}=\Omega\hat{\theta}$. The induced magnetisation follows $\boldsymbol{\Omega}\times\hat{h}\propto$

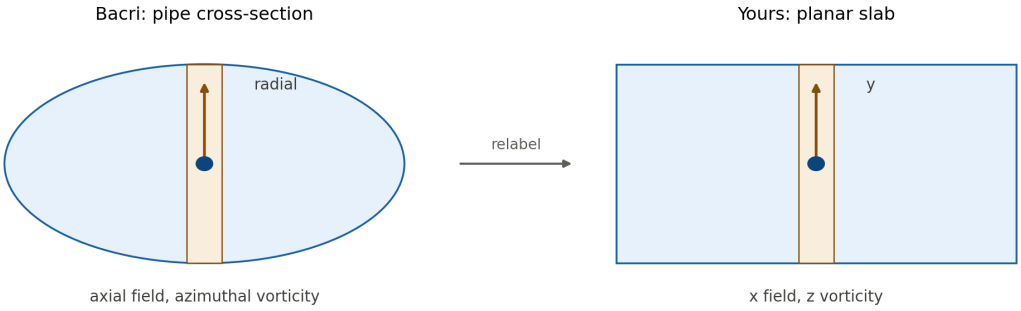


Figure 2: A fluid element in the pipe sees the same local arrangement as one in the slab. The shaded strip is the plane the magnetisation lives in; the dot marks the vorticity pointing out of that plane.

$\hat{\theta} \times \hat{x} = \hat{r}$ — radial. So at each point the magnetisation lives in the two-dimensional (x, r) plane with the vorticity perpendicular to it: exactly the slab configuration under $x \to x$, $r \to y$, $\theta \to z$.

(c) The geometry cancels out of the measurement. The two geometries obey different flow laws — $Q \propto \Delta p\, R^4/\eta$ for a pipe, $Q = \rho S F/2\eta$ for the slab — but the reported quantity is a *ratio* taken in one and the same apparatus:

$$g = \frac{2}{3\varphi_p}\left(\frac{\eta}{\eta_0} - 1\right) = \frac{2}{3\varphi_p}\left(\frac{Q_0}{Q} - 1\right), \tag{21}$$

since $\eta/\eta_0 = Q_0/Q$ at fixed driving. The R^4 cancels; so do S, ρ and F. The geometric factor is not approximately negligible — it is algebraically absent from the observable.

6. Where the argument stops working

Condition (C3) — the field must lie in the plane — is the one that cannot be relaxed. Suppose the field is tilted so that $H_z \neq 0$. Then in

$$\big[\mathbf{M} \times (\mathbf{M} \times \mathbf{H})\big]_z = M_z(\mathbf{M}\cdot\mathbf{H}) - H_z|\mathbf{M}|^2 \tag{22}$$

the first piece still vanishes ($M_z = 0$) but the second does not: $\dot{M}_z \propto -H_z|\mathbf{M}|^2 \neq 0$. The magnetisation immediately picks up an out-of-plane component and the planar description fails at once.

Physically: a field with a z-component exerts a torque with in-plane components, which tips the particle moments out of the plane. The symmetry that kept them there is gone.

A consequence worth knowing. Planar flow always has $\boldsymbol{\Omega}$ along $\hat{z}$, and (C3) puts $\mathbf{H}$ in the plane, so the angle β between $\mathbf{H}$ and $\boldsymbol{\Omega}$ is 90° *always*. The stationary-field result $\Delta\eta \propto \sin^2\beta$ therefore sits permanently at its maximum. Consequences:

(a) rotating $\mathbf{H}$ *within* the plane does not change $\Delta\eta$ at all;

(b) the case $\mathbf{H} \parallel \boldsymbol{\Omega}$, where $\Delta\eta = 0$ because aligned moments spin freely about the field, cannot be represented;

(c) McTague's result $\Delta\eta_{\parallel} = 2\Delta\eta_{\perp}$ cannot be tested.

These need a three-dimensional solver, not a modified two-dimensional one.

7. The fluid is itself magnetisable

So far **H** has been whatever magnetostatics delivers. We now ask what it actually delivers. Note first that this changes nothing about the planar reduction — §2.3 showed the field solver preserves the planar structure. What it changes is the *size* of **H**, and therefore of every magnetic effect.

7.1 How strongly does the fluid magnetise?

The susceptibility is $\chi = |\mathbf{M}|/|\mathbf{H}|$. For the present parameters, with $M_s = 3\eta\varphi_p\xi_0/(\mu_0\tau_B H_0)$,

$$\chi = \frac{M_s L(\xi_0)}{H_0} = 0.15\,\xi_0 L(\xi_0)\,: \tag{23}$$

ξ_0	1	2.5	5	10	20
χ	0.05	0.16	0.51	1.34	2.85

At weak fields the fluid is magnetically almost invisible. At strong fields it magnetises nearly three times as hard as the field driving it — it has become a magnet in its own right.

7.2 A magnetised fluid changes the field inside itself

Take x-invariant plane shear with $\mathbf{H}_{\text{app}} = H_x\hat{x}$. The flow tilts the moments and produces a transverse magnetisation $M_y \propto \Omega(y)$. For Poiseuille flow Ω is zero at the centreline and largest at the walls, with opposite signs on the two sides, so M_y varies across the channel and

$$\nabla\cdot\mathbf{M} = \partial_y M_y \neq 0. \tag{24}$$

Magnetic poles therefore appear: north on one wall, south on the other. From (7) with $\partial_x = 0$,

$$\partial_y\big(H_y + M_y\big) = 0 \implies H_y + M_y = \text{const} \xrightarrow[\text{about the centreline}]{M_y \text{ odd}} \boxed{H_y = -M_y} \tag{25}$$

Substituting into the torque:

$$\big(\mathbf{M}\times\mathbf{H}\big)_z = M_x H_y - M_y H_x = -M_y\big(M_x + H_x\big) = -M_y H_x(1+\chi). \tag{26}$$

In words: the fluid's own field adds to the applied one and the whole effect is multiplied by $(1+\chi)$ — a factor running from 1.05 to 3.85 across the range of ξ_0 swept in Fig. 3. Because it is not constant, it distorts the *shape* of the curve and cannot be removed by rescaling.

Note on the pipe. It is sometimes supposed that the cylindrical geometry avoids this. It does not. As shown in §5, the induced magnetisation in the pipe is *radial*, and with $M_r \propto \Omega(r) \propto r$ we get $\nabla\cdot\mathbf{M} = r^{-1}\partial_r(rM_r) = \text{const} \neq 0$. Solving (7) gives $r(H_r+M_r) = \text{const}$, and finiteness at $r = 0$ forces $H_r = -M_r$ — the same result as the slab. The demagnetising field is present in both geometries. Its absence from the source paper is an approximation, not a property of the apparatus.

7.3 Practical consequences

1. **Reproducing Fig. 3 requires leaving it out.** Since the source theory omits the self-field, a solver that includes it *must* disagree with Fig. 3, by roughly $(1 + \chi)$. Matching Fig. 3 is a test of the solver against that theory and requires adopting the same omission. Keeping a switch for this is worthwhile: it is the only closed-form check available.
2. **The Kelvin force returns.** In a uniform field $\mu_0(\mathbf{M}\cdot\nabla)\mathbf{H} = 0$ identically. A self-field makes $\mathbf{H}$ non-uniform and this term comes back — it is the force behind all magnetic pumping.
3. **Interfaces need it.** At a free surface $\mathbf{M}\cdot\hat{n} \neq 0$ creates surface poles. This *is* the mechanism of the normal-field (Rosensweig) instability: without solving (7) that physics is absent altogether, not merely inaccurate.
4. **Validation.** A circular cylinder in a uniform field has a closed-form solution and is the standard first test for a magnetostatic solver.

8. Limitations of the model, and what would fix them

The reduction of §3 is exact *for the model*. The model itself carries the following approximations. None is caused by working in two dimensions, and none is repaired by re-deriving anything — they are fixed by adding physics.

8.1 The field description

(L1) The demagnetising field is neglected. As in §7. The source equations treat $\mathbf{H}$ as the applied field and never solve (7). Size of the error: up to $(1 + \chi) \approx 3.9$ at $\varphi = 0.2$. *Fix:* solve for the magnetic scalar potential alongside the flow.

(L2) No spin diffusion. Couple-stress terms that smooth rapid spatial variation of $\boldsymbol{\omega}_p - \boldsymbol{\Omega}$ are dropped. Harmless in smooth shear flow; questionable near obstacles, stagnation points and interfaces. *Fix and benchmark:* Rinaldi & Zahn, Phys. Fluids **14**, 2847 (2002) give plane-Poiseuille and planar-Couette solutions *with* spin diffusion, including analytic velocity, spin and vorticity profiles — directly usable in the present geometry.

8.2 The particle description

(L3) A dilute result used at $\varphi = 0.2$. The coefficient $6\eta\varphi$ comes from Stokes drag on an *isolated* sphere; Shliomis notes that the rotational viscosity and Einstein's $\frac{5}{2}\eta\varphi$ are of the same order, confirming it is a dilute-suspension result. At $\varphi = 0.2$ hydrodynamic and dipolar interactions between particles are not negligible. *Fix:* relaxation equations derived for concentrated, interacting ferrofluids, which recover Martsenyuk–Raikher–Shliomis as their dilute limit.

(L4) Chain formation excluded. Experiments on well-characterised cobalt ferrofluids show the zero-field viscosity rising by factors of 10–100 under a field, accompanied by chain-like aggregation seen by neutron scattering under shear. The present model cannot produce this at all. *Fix:* kinetic chain models; see Odenbach (ed.), *Magnetoviscous Effects in Ferrofluids*, Lecture Notes in Physics m71 (2002).

(L5) Rigid dipoles only. The moment is assumed to reorient only by rotation of the whole particle. Particles that relax by the Néel mechanism — the moment flipping inside a stationary particle — exert no torque on the carrier and contribute no rotational viscosity. Correct for the

paper's Co-ferrite ($K = 2\times10^6\,\mathrm{erg/cm^3}$, moments frozen up to $10^4\,\mathrm{Oe}$); wrong for fine magnetite. *Fix:* add a Néel channel, $\tau_N^{-1} = f_0 e^{-KV/k_BT}$, with $\tau^{-1} = \tau_B^{-1} + \tau_N^{-1}$.

(L6) One particle size. A single τ_B and a single moment m. Since $\tau_B \propto d^3$, a realistic log-normal distribution spreads the response over orders of magnitude in frequency and blurs the sharp features of Fig. 3. *Fix:* superpose over a measured size distribution.

8.3 Specific to the published figure

(L7) Truncation at first order in $\Omega\tau_B$. Bacri Eqs. (7)–(8) discard terms of order $(\Omega\tau_B)^2$. The full equation (6) does not, and is known to remain accurate even for $\Omega\tau \gg 1$, whereas the simpler Debye closure is reliable only for $\Omega\tau \lesssim 1$. A simulation integrating (6) therefore *exceeds* the accuracy of the reference curve, and the small residual difference measures the truncation rather than the solver.

(L8) Which magnetisation equation. This was disputed. Felderhof (Phys. Rev. E **62**, 3848 (2000)) proposed an alternative; Shliomis rebutted (Phys. Rev. E **64**, 063501 (2001)). It was settled numerically in favour of Martsenyuk–Raikher–Shliomis by rotational Brownian-dynamics simulation (Soto-Aquino & Rinaldi, Phys. Rev. E **82**, 046310 (2010)) and by direct integration of the Fokker–Planck equation. Equation (6) is the surviving closure.

8.4 Structural assumptions

(L9) Incompressible, isothermal, non-conducting carrier; no thermomagnetic coupling (no pyromagnetic term, no temperature dependence of M_s or η). **(L10)** No magnetophoretic migration of particles, so φ stays uniform; in a field gradient particles drift and φ becomes a transported field. **(L11)** Rotational inertia of the particles neglected ($\tau_s \sim 10^{-11}\,\mathrm{s}$) — entirely safe.

8.5 Idealisations of the planar geometry

Not model errors, but worth stating when reporting results:

- Demagnetising factors are those of a slab infinite in z: $N_y = 1$, $N_x = N_z = 0$, not those of a rectangular duct.
- Obstacles are infinite cylinders, not spheres, so drag coefficients are cylinder values.
- Surface patterns become stripes: the Rosensweig instability forms a hexagonal lattice in the plane of a real film, whereas a planar cross-section gives a periodic ridge array. The onset threshold and wavelength are right; the pattern symmetry is not.

Summary. With the field in the plane of the flow, every term of the ferrohydrodynamic equations has zero out-of-plane component, so the planar computation reproduces the three-dimensional dynamics exactly. The only condition that cannot be relaxed is that the field stay in the plane. Making the medium magnetisable does not affect this — the field solver preserves the planar structure — but it multiplies every magnetic effect by $(1 + \chi)$, which is not constant across the parameter range. The remaining limitations (L1)–(L11) belong to the model rather than to the reduction.